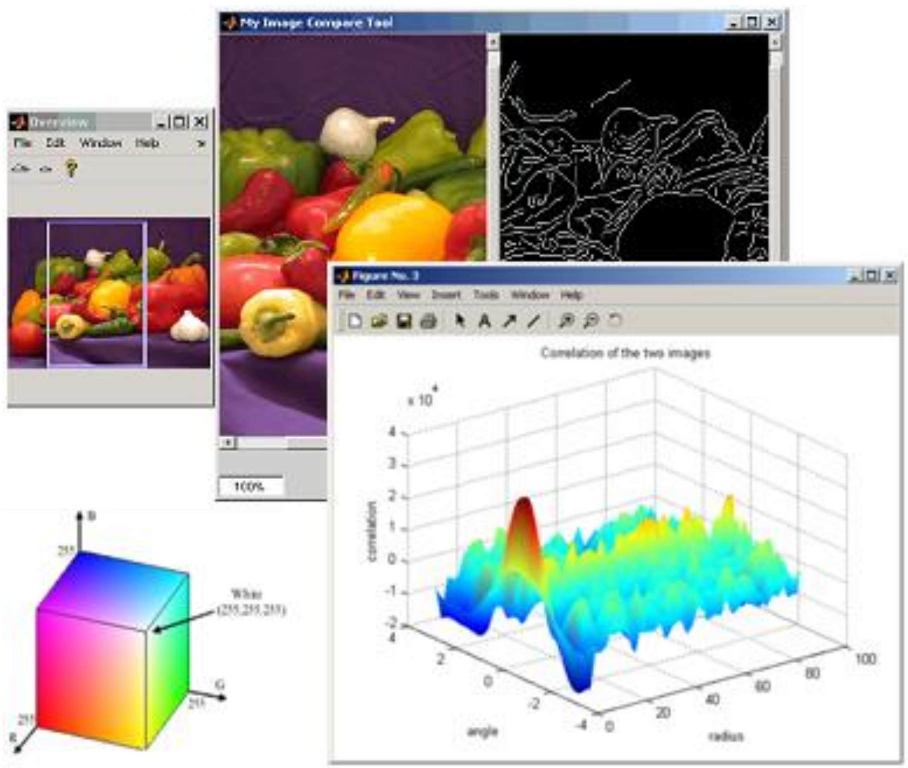
Digital Image Processing

LECTURE 2: INTRODUCTION



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Assistant Professor
ECE Department
GLA University Mathura

Introduction and Fundamentals

Introduction

When can we Object be Imaged?

Introduction

Introduction

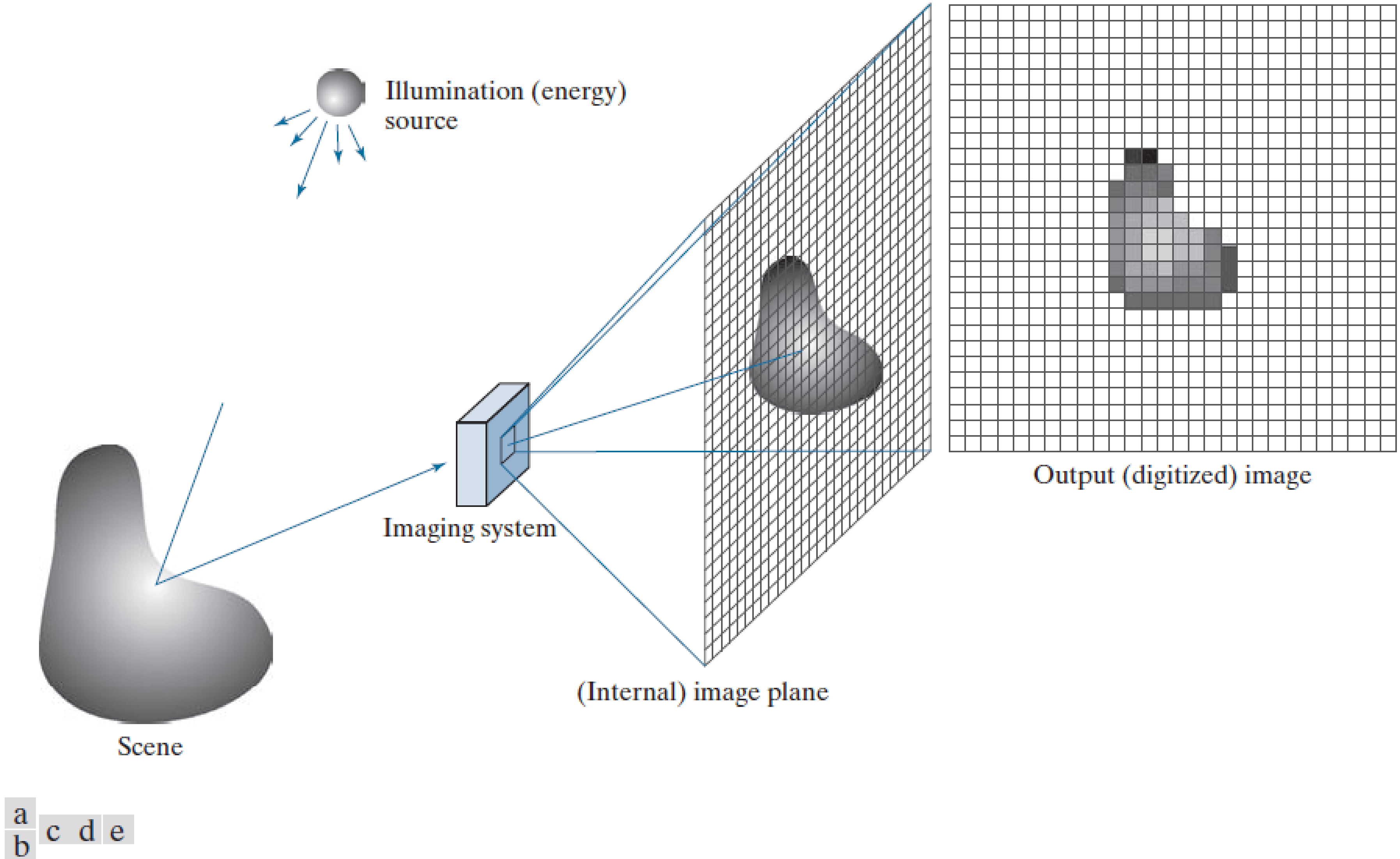


FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Introduction

$$f(x, y)$$

Where

x, y : represents the spatial coordinates

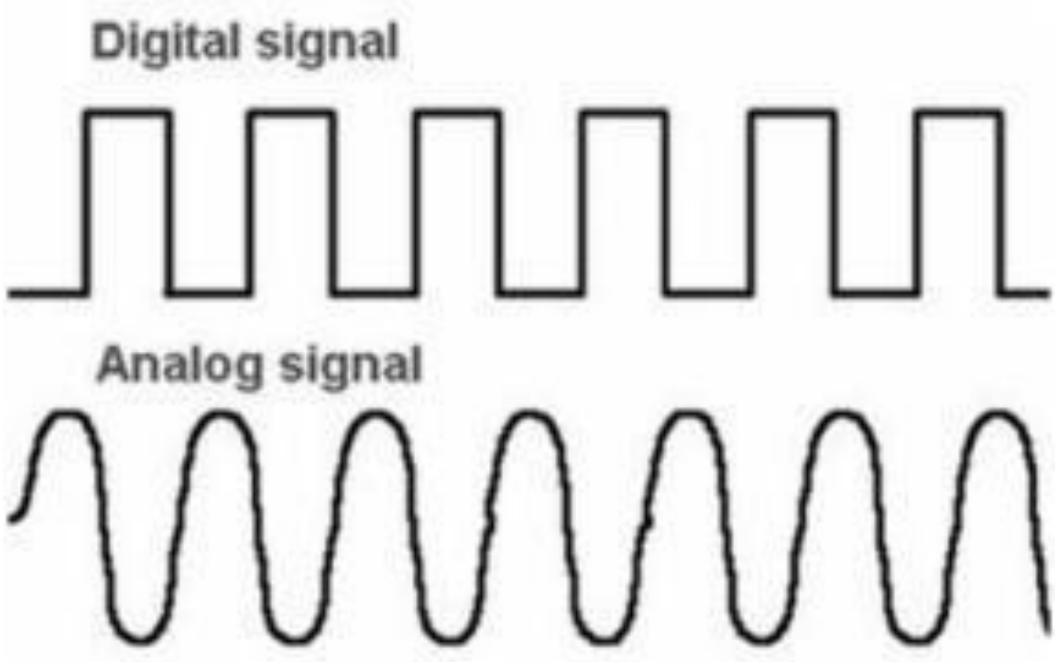
f: represents the amplitude/intensity

Image

Analog Image
(if x and y are continuous)
f is also continuous

Digital Image
(if x and y are discrete)
f is quantized

Analog signal Vs Digital Signal



- **Analog Signal:** An analog signal is a continuous signal that varies smoothly over time.

- **Digital Signal:** Digital signals are not continuous; instead, they are sampled at specific intervals

Introduction

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Where

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Image

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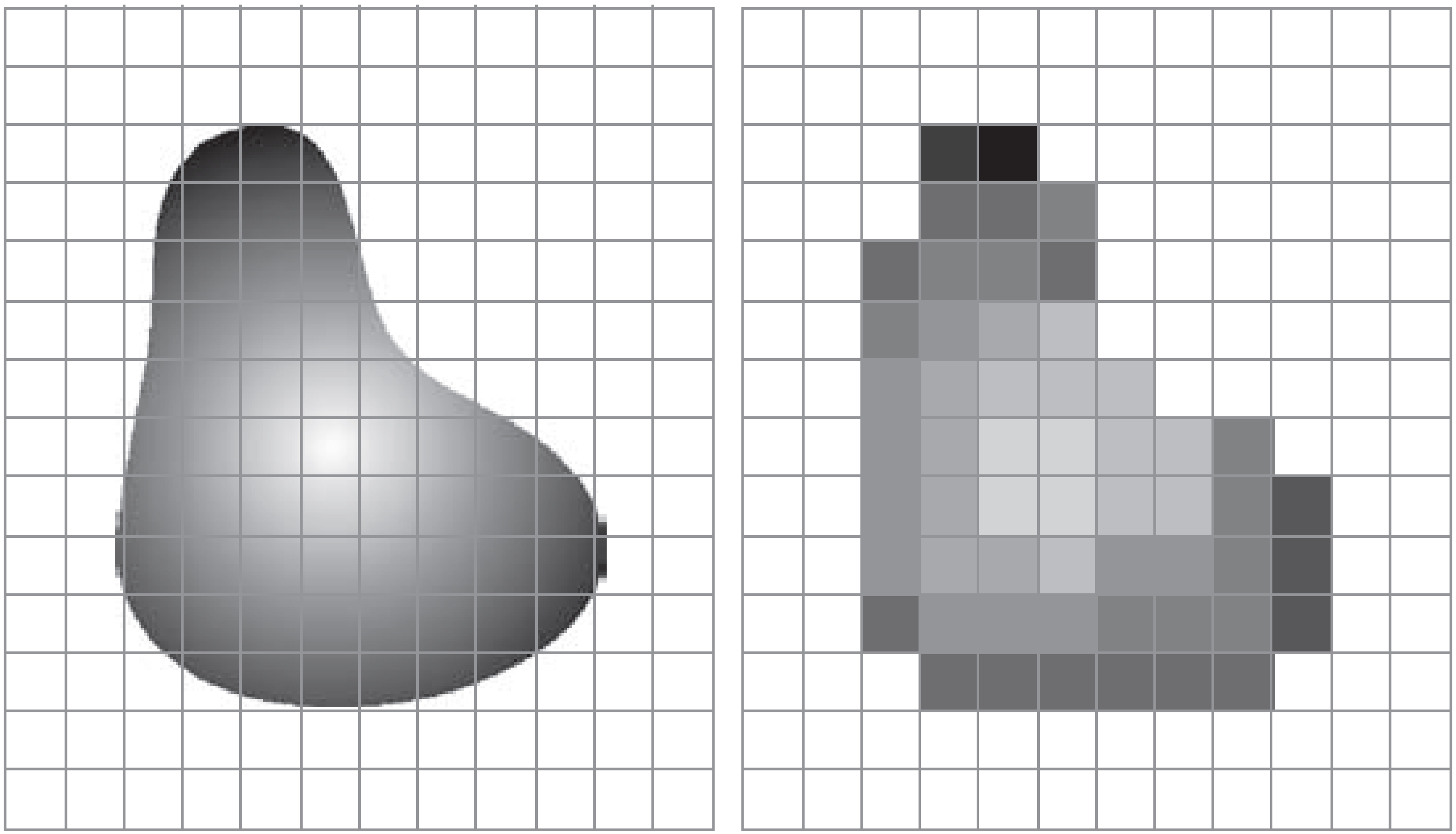
(if x and y are continuous)
f is also continuous

Digital Image
(if x and y are discrete)
f is quantized

Introduction



a b

FIGURE 2.17
(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

What is an Image?

Projection of 3D scene in 2D plane is called as an “Image”.

What is an Image ?

- An image is a visual representation of scene.
- An image is an item that depicts visual perception such as a photograph.
- An image is a 2-D representation of a 3-D scene.
- It plays an important role in various applications areas such as research and technology.
- In the context of signal processing, an image is a collection of distributed amplitude of color(s).

Analog Image

- An analog image is characterized by a continuous variation in tone, where each tone represents a continuous variation of the scene being captured.
- This definition is commonly applied in the context of remote sensing, where analog images are produced by photosensitive chemicals in photographic film.

Aerial Photography

Examples:

X-ray Imaging

Painting

CRT monitor

Digital Image

- This representation involves dividing the image into small, discrete units called pixels, which are the smallest individual elements in the image.
- Each pixel is holding quantized values that typically denote color and brightness at specific points.
- A digital image is essentially a representation of a real image that is converted into a format that can be stored and processed by a digital computer.
- Digital Image Formats:• RAW • BMP • GIF • JPG • PNG • TIFF • HDF

1	1	1	1	1	1	1	1	1	1
1	0	0	0	1	1	0	0	0	1
1	1	0	1	1	1	1	0	1	1
1	1	0	1	1	1	1	0	1	1
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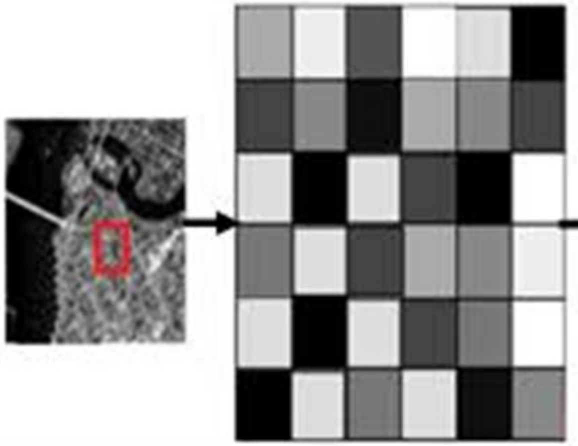


170	238	85	255	221	0
68	136	17	170	119	68
221	0	238	136	0	255
119	255	85	170	136	238
238	17	221	68	119	255
85	170	119	221	17	136

Analog Image vs Digital Image

❑ Advantages of Digital image over Analog images

- It is easy to post-process the image. Small corrections can be made in the captured image using software.
- It is easy to store the images in the digital memory
- It is possible to transmit the image over network. So, sharing an image is quite easy.
- A digital image does not require any chemical process. So, it is very environment friendly, as harmful film chemicals are not required ot used.
- It is easy to operate a digital camera.

Analog Image vs Digital Image

❑ Disadvantages of digital images:

- Initial cost,
- Problems associated with sensors such as high-power consumption and potential failure
- Security issues associated with the storage and transmission of digital images.

Digital Image

- A digital image is a **binary representation of visual information**
- A digital image is a representation in a **two-dimensional space** as a **finite set of digital values**, called **picture elements** or **pixels**.

❑ Digital Image Representation:

- An image can be defined as a 2D signal that varies over the spatial coordinates x and y, and can be written mathematically as f(x,y)

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & .. & .. & f(0,Y-1) \\ f(1,0) & f(1,1) & f(1,2) & .. & .. & f(1,Y-1) \\ .. & .. & .. & .. & .. & .. \\ f(X-1,0) & f(X-1,1) & f(X-2,2) & .. & .. & f(X-1,Y-1) \end{bmatrix}$$

Digital Image Representation

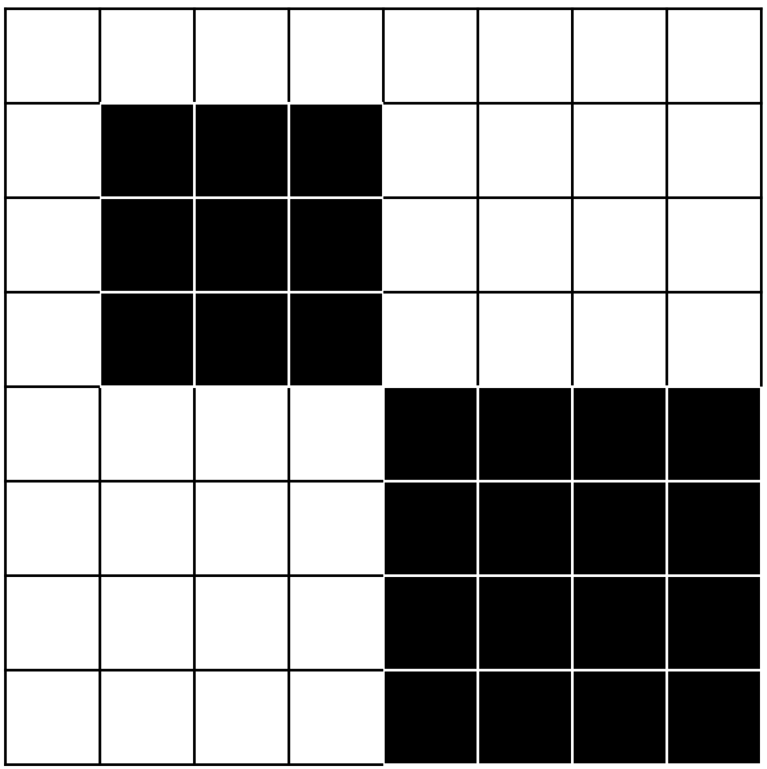
- f(x,y) is divided into X rows and Y columns
- At intersection of rows and columns, pixels (picture element) are present.
- The value of the function f(x,y) at every point indexed by a row and a column is called grey value or intensity of the image.
- **Vertical Resolution:** The number of rows in a digital image is called vertical resolution
- **Horizontal Resolution:** The number of columns is called horizontal resolution.
- **Dimension of an image:** The number of rows and columns describes the dimension of the image.

Image Representation

How to Computer Understand Images?

Image Representation

Binary Image Representation



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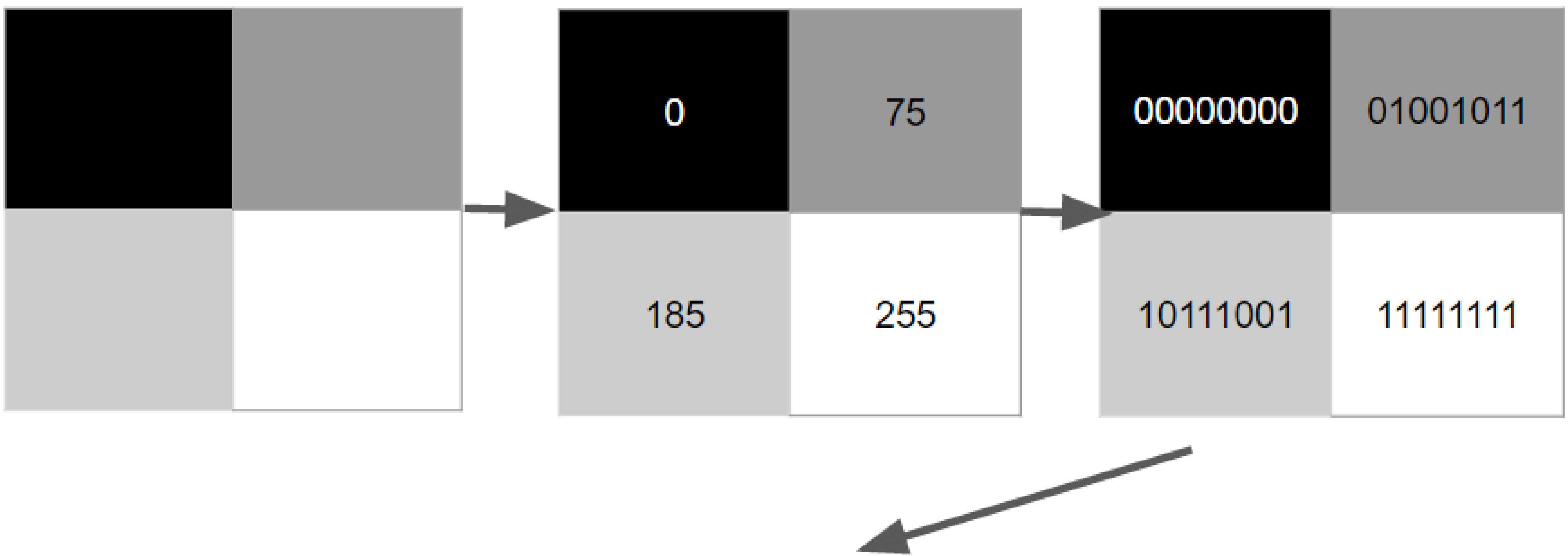
1	1	1	1	0	0	0	0
1	1	1	1	0	0	0	0
1	1	1	1	0	0	0	0

Image Representation

Gray Image Representation



00000000 01001011 10111001 11111111

Image Representation

- **Binary images:** In binary images, the pixels assume a value of 0 and 1. The binary image is created from a grey scale image using a thresholding process.

- **Gray scale images:** In Grey scale images, they have many shades of grey between black and white.

Original Gray Image A

Output Binary Image B

Image Representation

Colour Image Representation

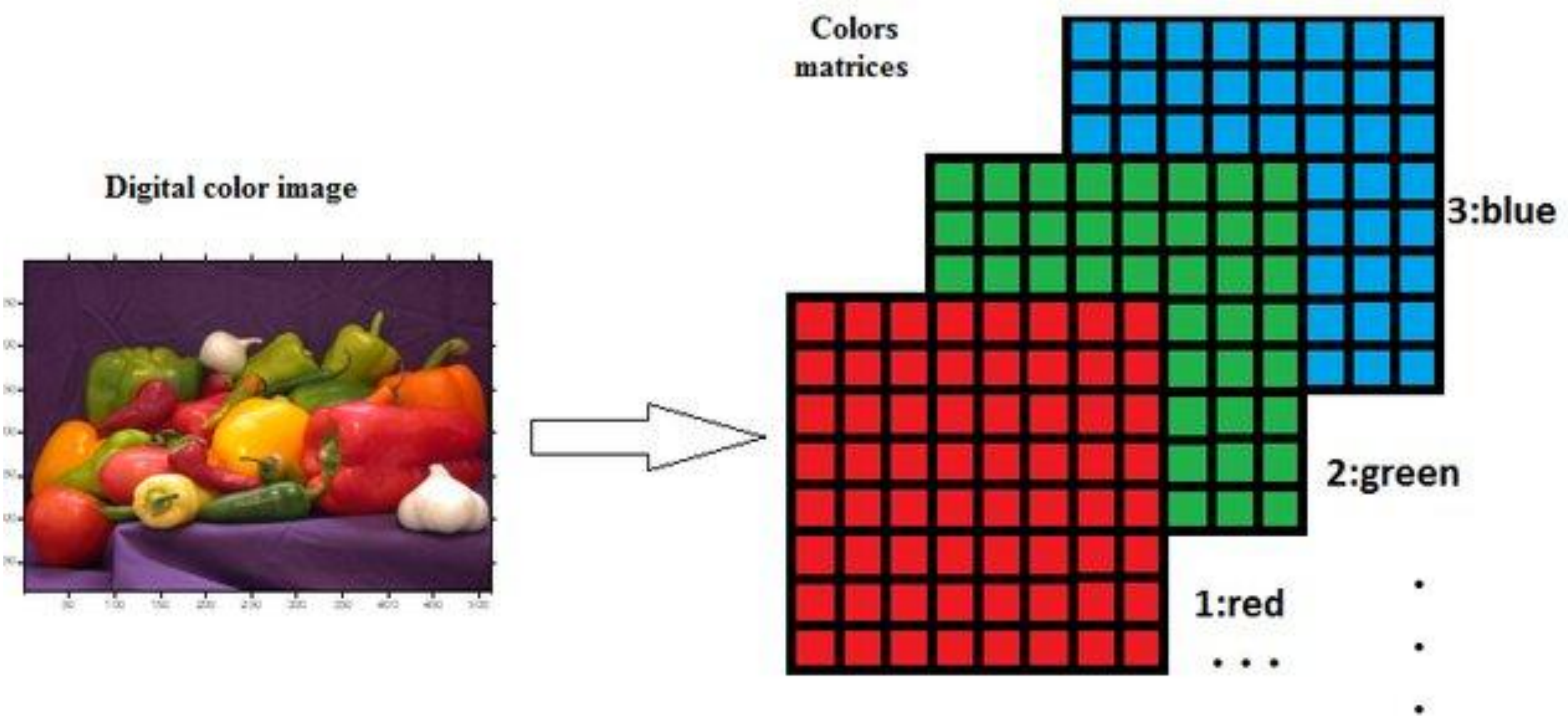
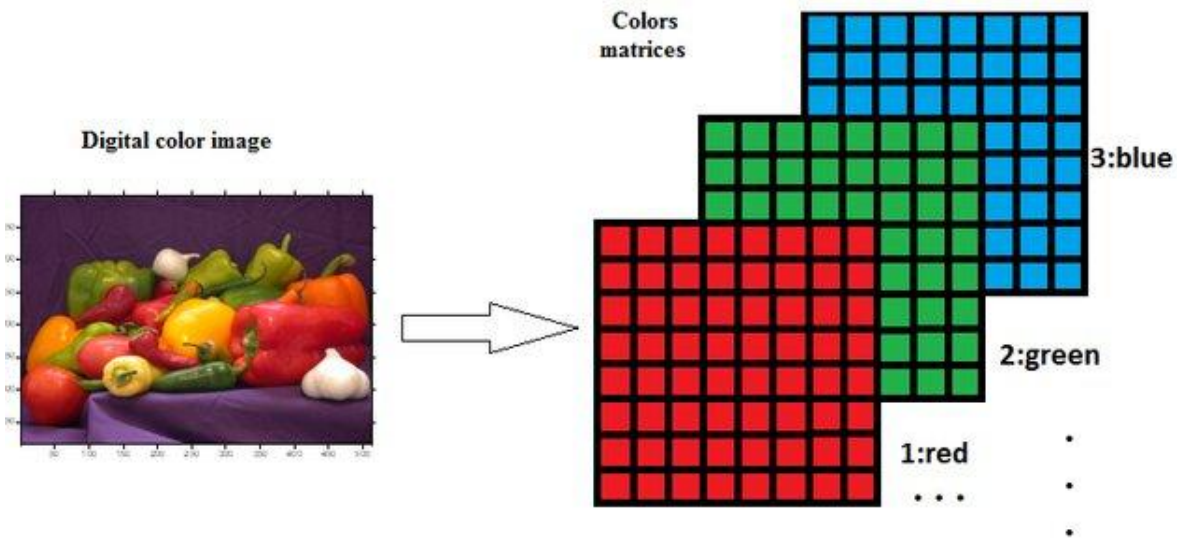


Image Representation



- Digital Image data can also have a third dimension called layers.
- Layers are the images of the same scene but containing different information.
- In multi-spectral images, layers are the images of different spectral ranges called bands or channel.
- For instance, a colour picture, taken by a digital camera is composed of three bands containing red, green and blue spectral information individually.

Image Representation

Colour Image Representation

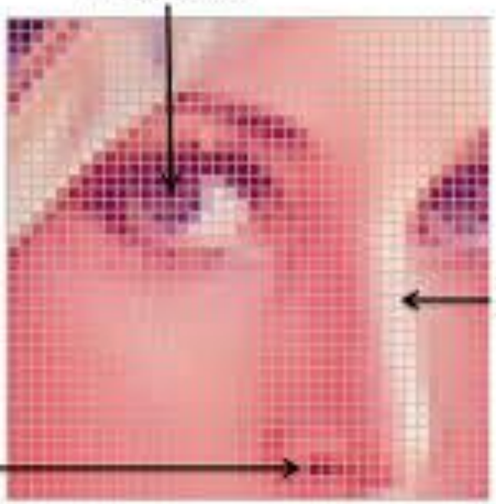


Image Representation

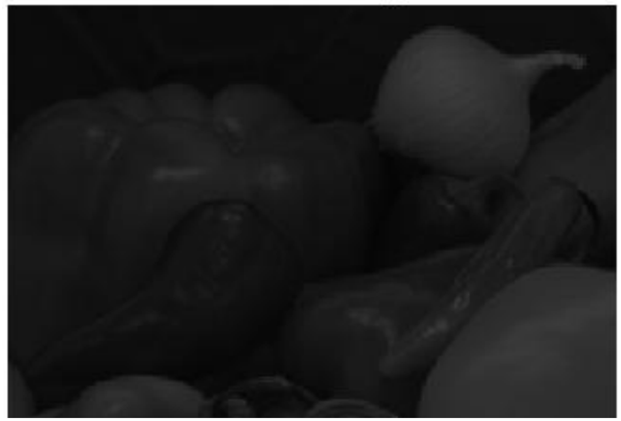
- **True colour images:** In true colour images, the pixel has a colour that is obtained by mixing the primary colours (Red, Green and Blue). Hence it is also called as three-band images.

- **Pseudo colour images:** In remote sensing applications, multi-band images or multi-spectral images are generally used. These images, which are captured by satellite, contain many bands. A typical remote sensing image may have a 3- 11 bands in an image. This information is beyond the human perceptual range. So, **colour is artificially added** to these bands and to increase operational convenience. These are called artificial colour or pseudo colour images.

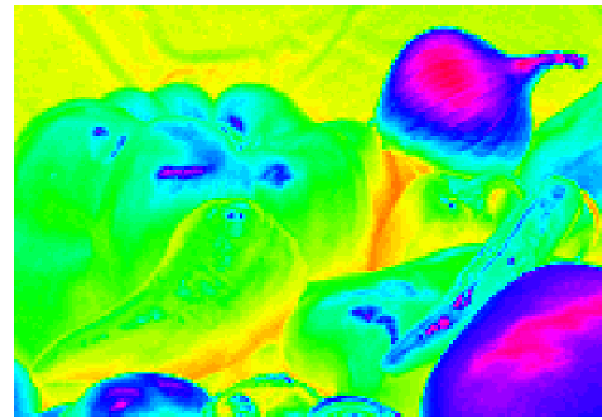
Pseudo Images

Original Colour Image

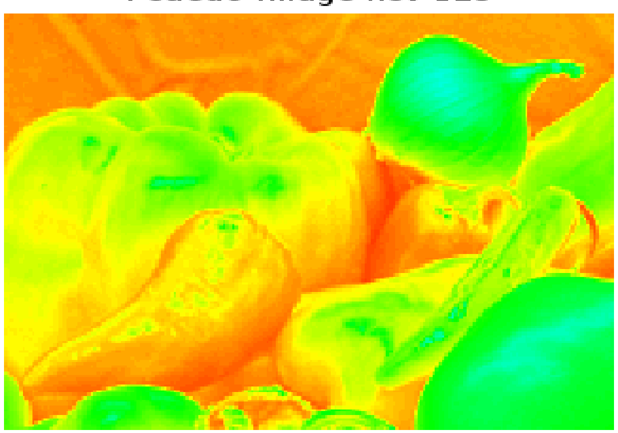
Indices Image



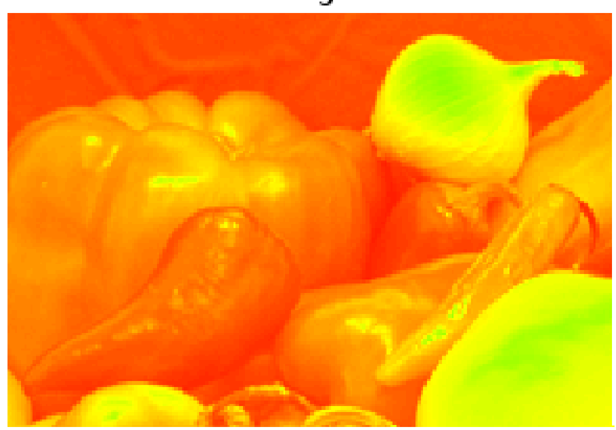
Psuedo Image hsv 64



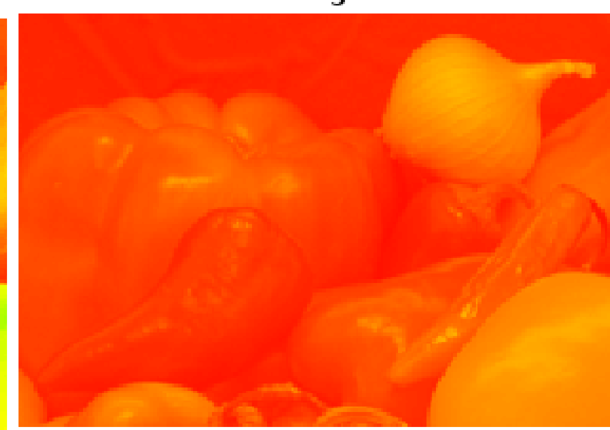
Psuedo Image hsv 128



Psuedo Image hsv 256



Psuedo Image hsv 512



Pseudo Image

Original Image A

Image B: ntsc representation of A

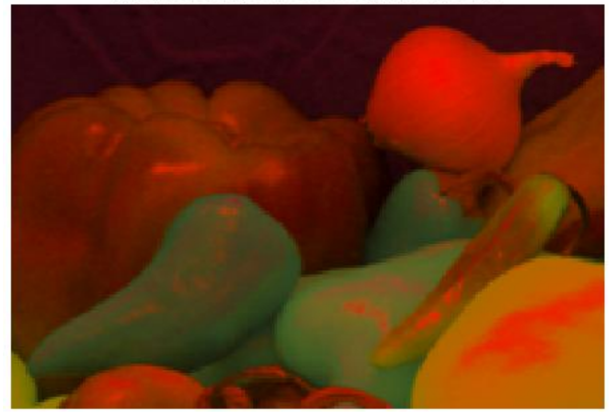


Image C: ycbcr representation of Image A

Image D: CIE 1976 XYZ representation of Image A

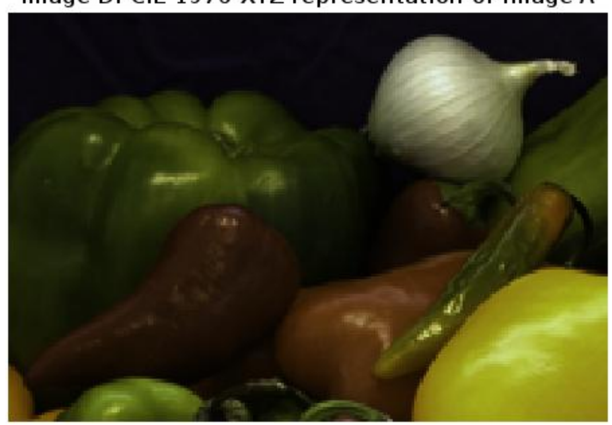


Image E: CIE 1976 lab representation of Image A

Digital Image Processing

Digital Image Processing means processing of images which are digital in nature by using a digital computer

Digital Image Processing

- Manipulation of digital images by means of a digital computer for representation in suitable form is known as Digital Image Processing.
- Digital image processing focusses on two major tasks:
  - Improvement of pictorial information for human interpretation.
  - Processing of Images data for storage, transmission and representation for autonomous machine perception.

Digital Image Processing

The image processing can be broken up into low-level, mid-level and high-level processes as follows:

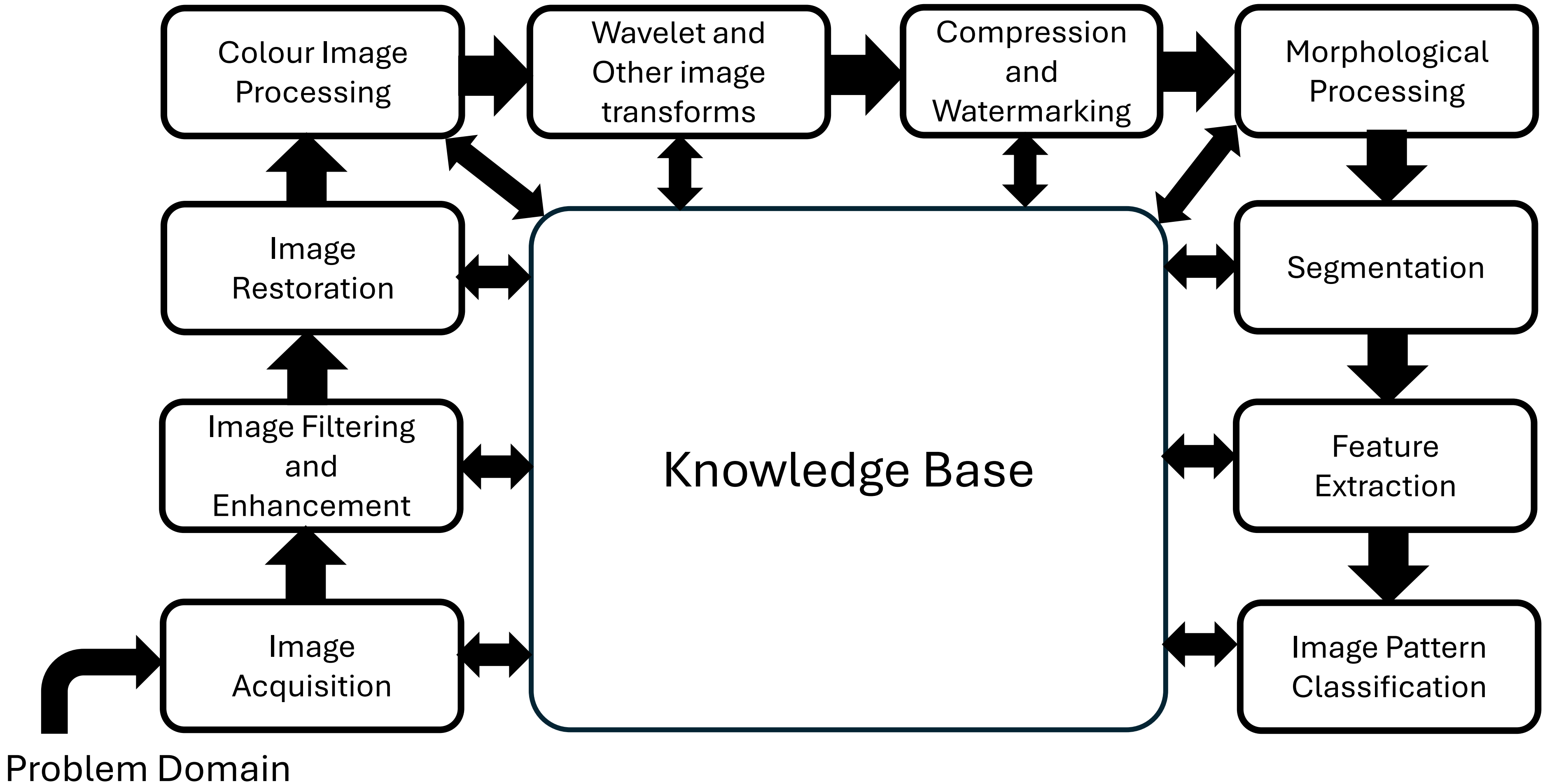
LOW LEVEL PROCESS	MID LEVEL PROCESS	HIGH LEVEL PROCESS
INPUT: Image	INPUT: Image	INPUT: Attributes
OUTPUT: Image	OUTPUT: Attributes	OUTPUT: Understanding
Examples: Noise removal, Image Sharpening	Examples: Object Recognition, Image Segmentation	Examples: Autonomous Navigation

Introduction

What we can do with Digital Images?

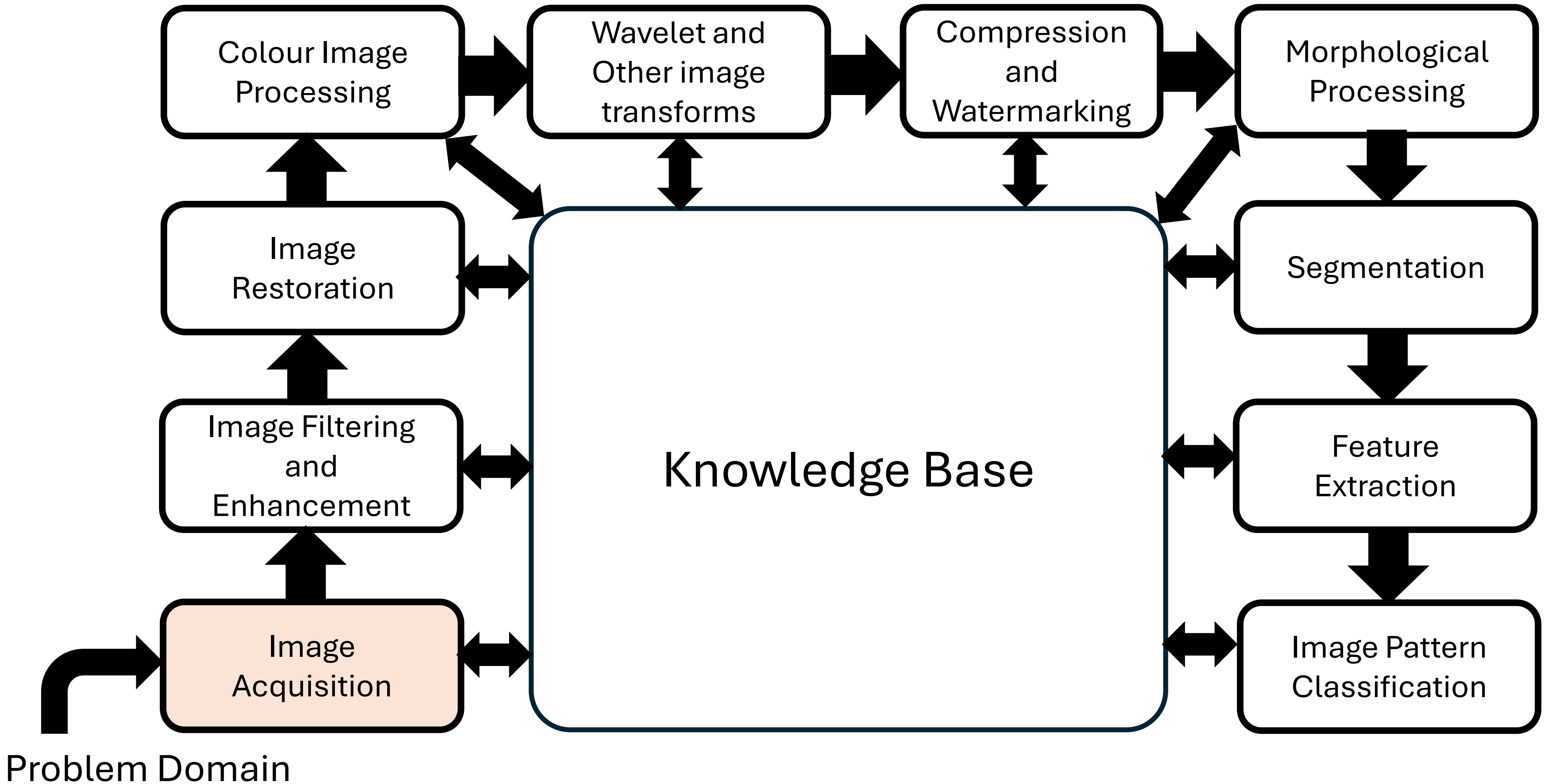
Fundamentals Steps in Digital Image Processing

Fundamental Steps in Image Processing



Fundamental Steps in Image Processing

Image Acquisition: This step aims to obtain the digital image of the object



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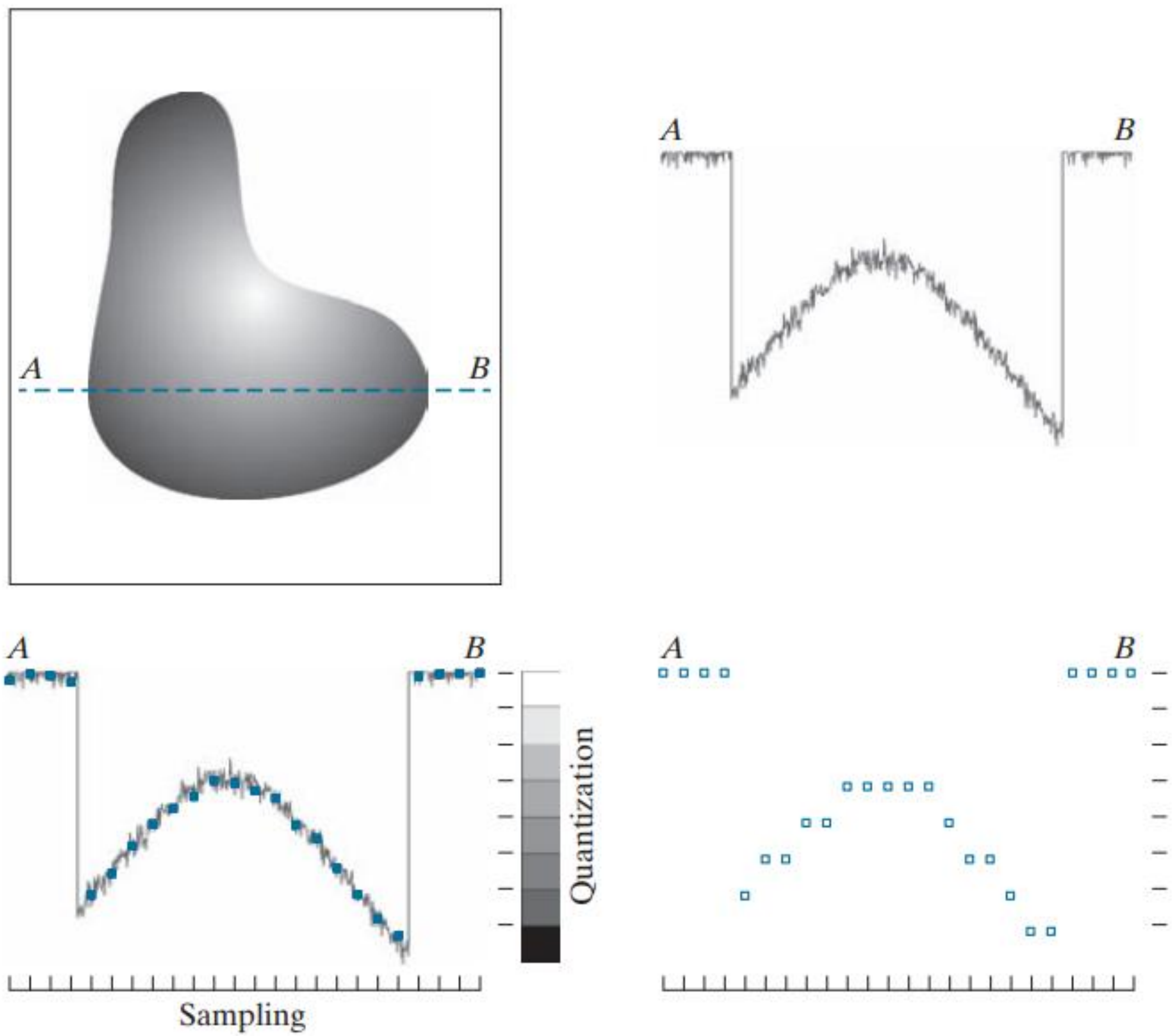
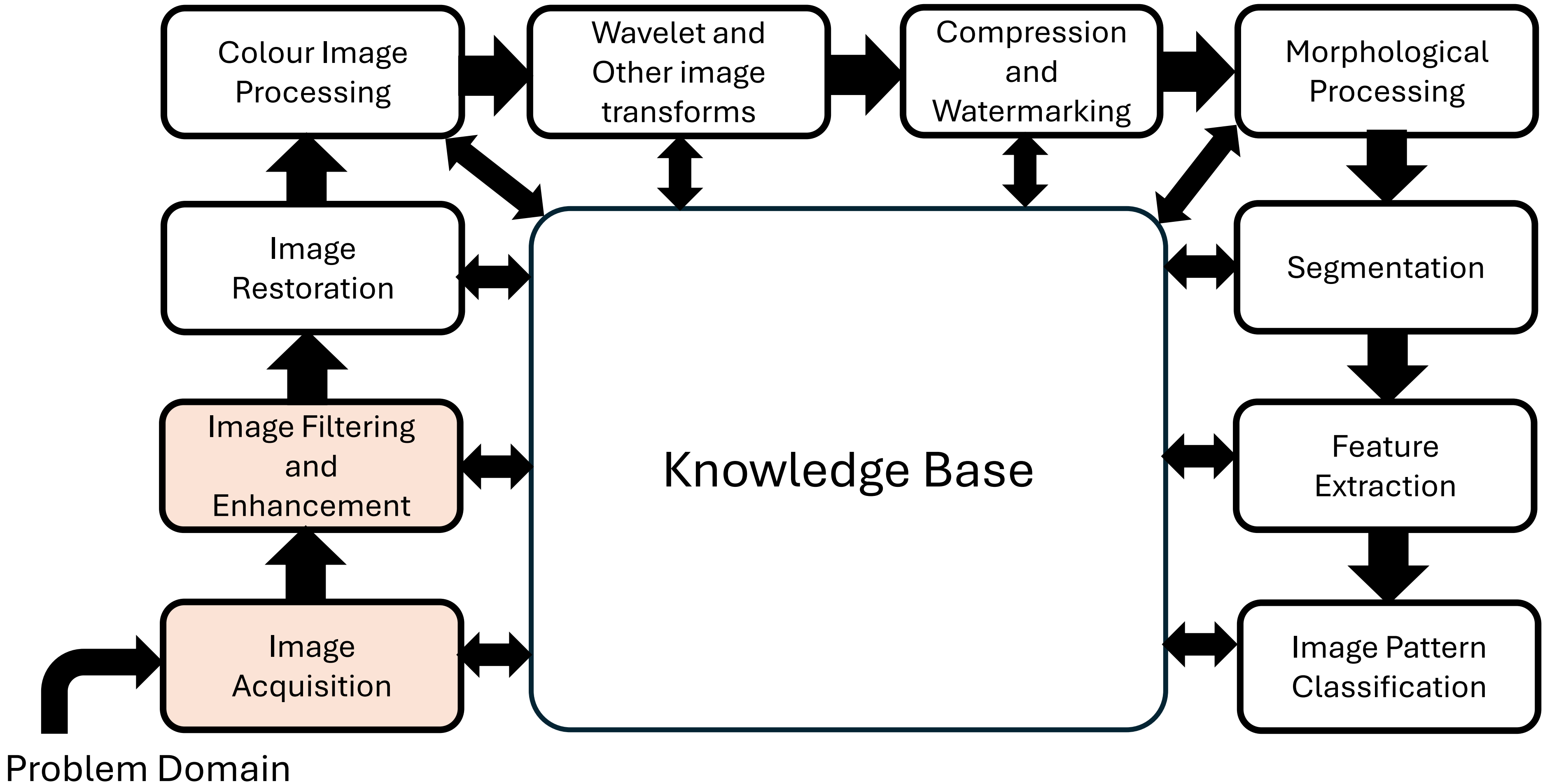


Fundamental Steps in Image Processing

Image Enhancement: This step aims to improve the quality of the image so that the analysis of the image is reliable.



Fundamental Steps in Image Processing

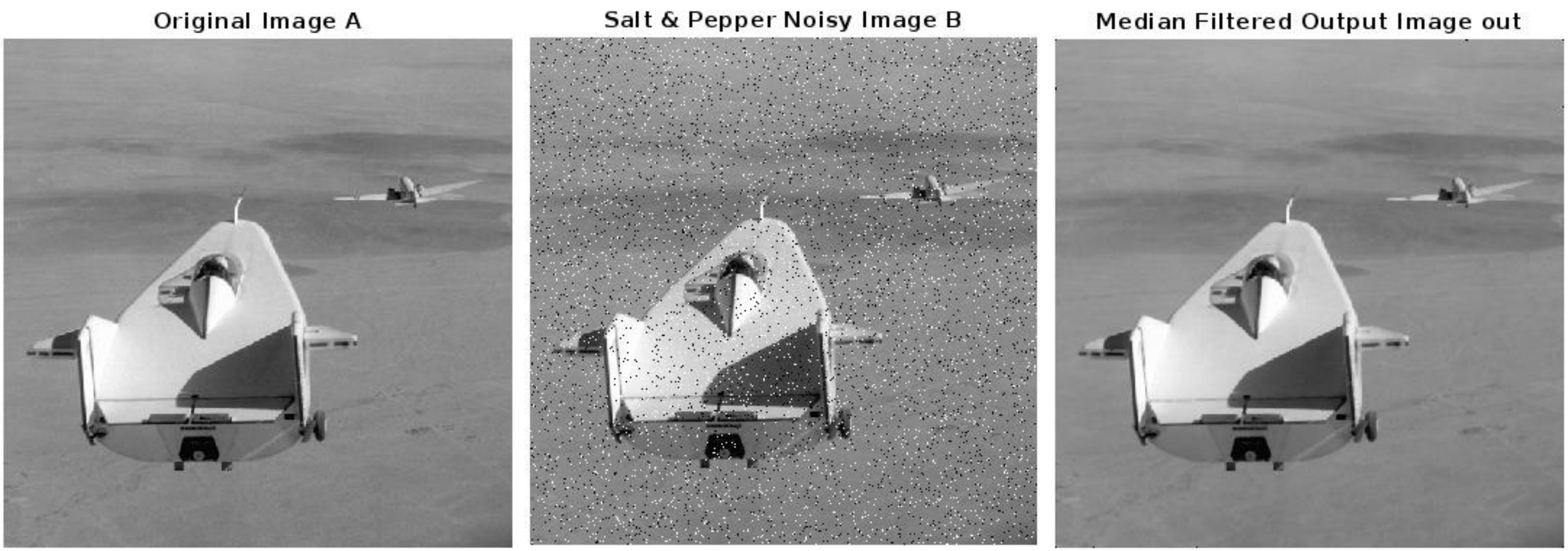
Image Enhancement: This step aims to improve the quality of the image so that the analysis of the image is reliable.

Original Image A

Enhanced Output Image B

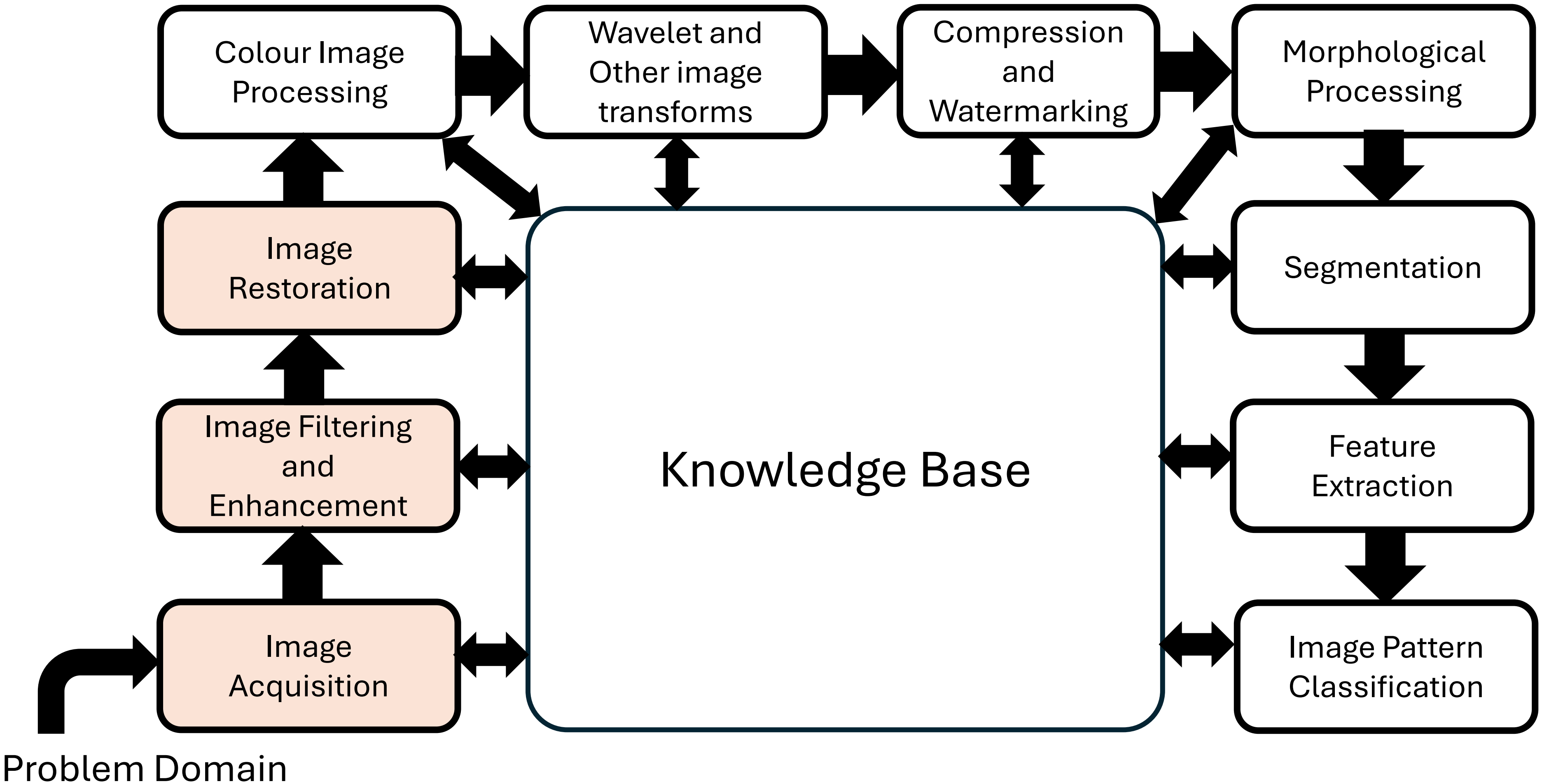
Fundamental Steps in Image Processing

Image Filtering: This step aims to remove the noise or sharpen the edges present in an digital Image.



Fundamental Steps in Image Processing

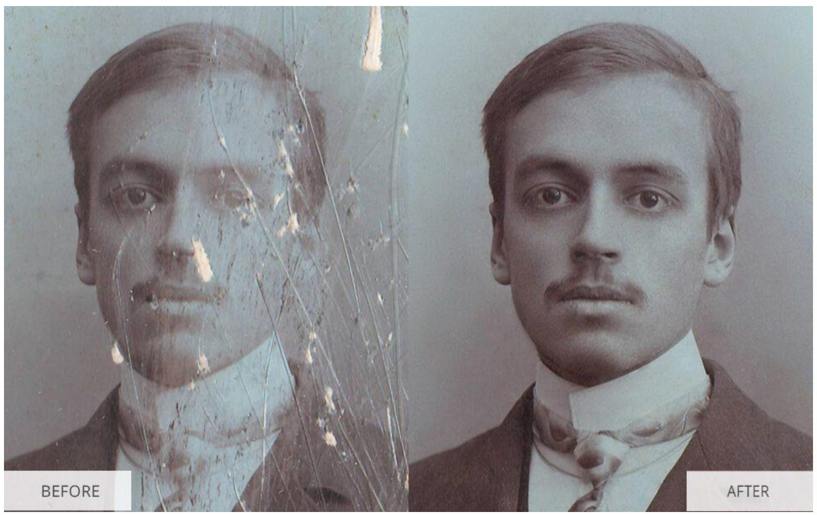
Image Restoration: It is an area that also deals with improving the appearance of an image.

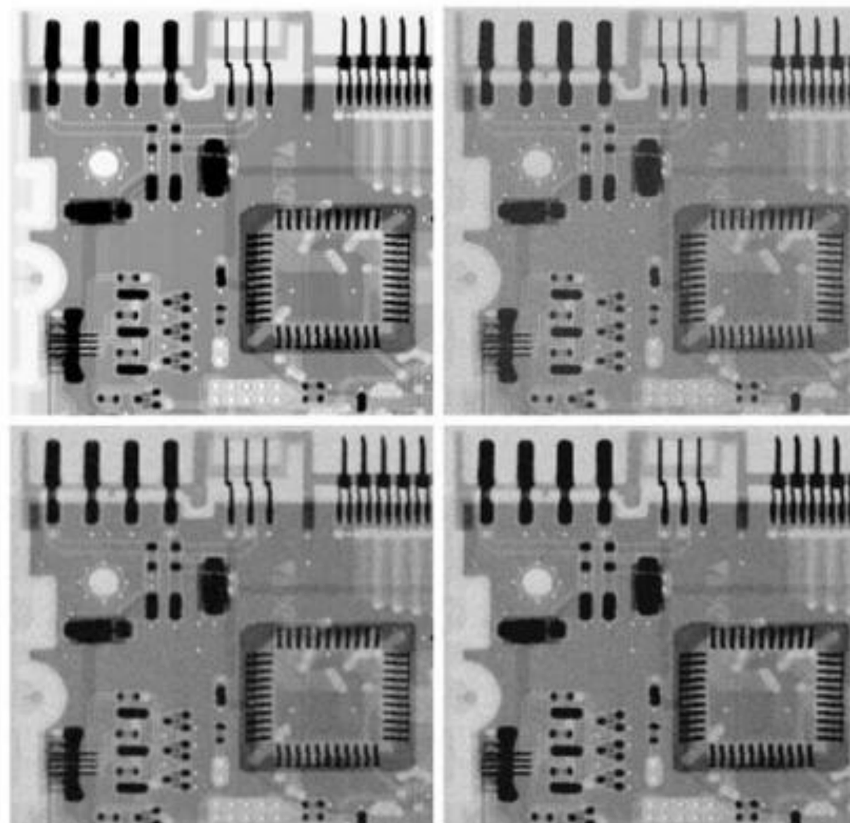


Fundamental Steps in Image Processing

Image Restoration:

- It is an area that also deals with improving the appearance of an image.
- Restoration attempts to recover an image that has been degraded by using a priori knowledge of degradation phenomenon.



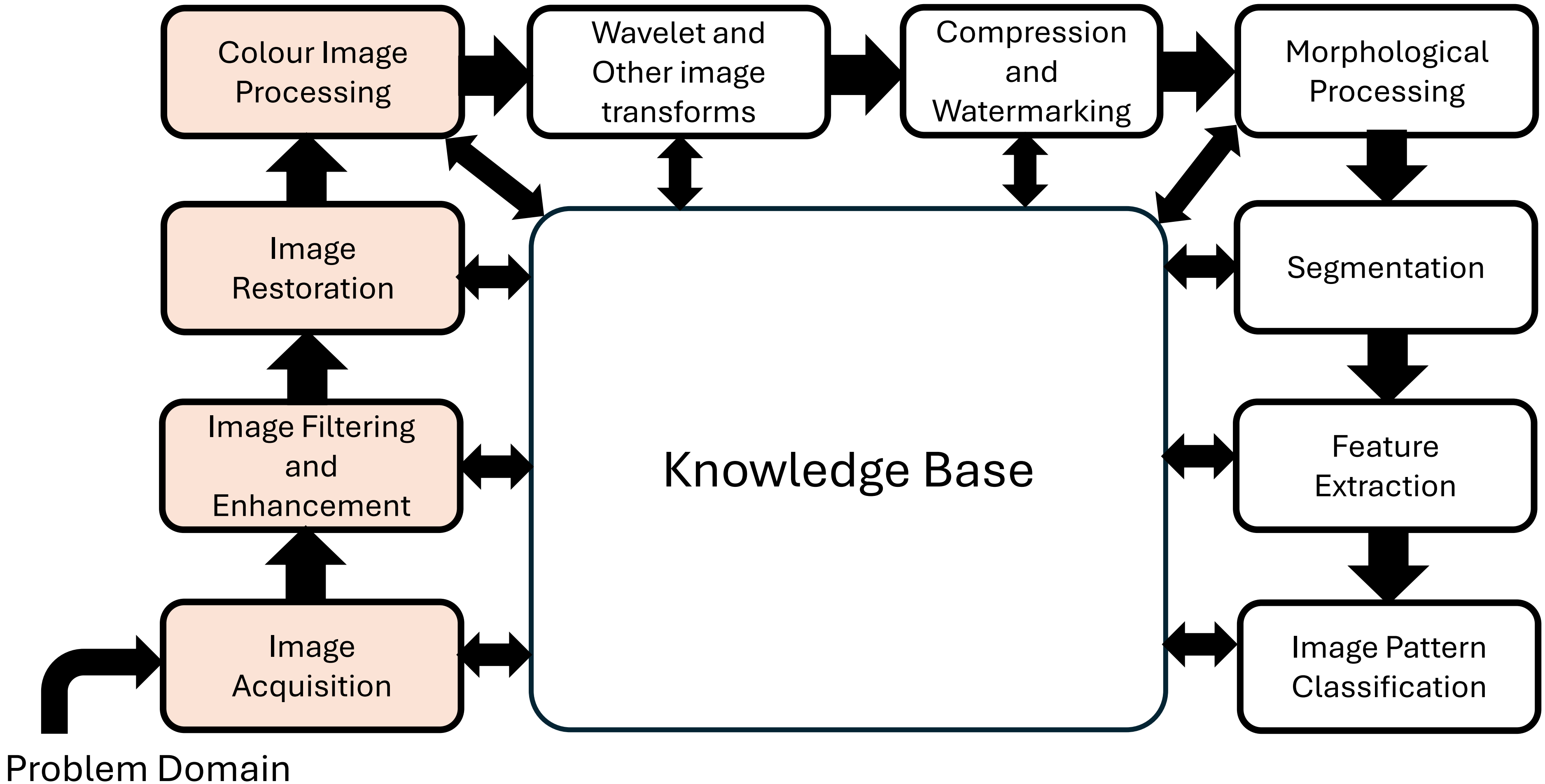


a b
c d

FIGURE 5.7 (a) X-ray image. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3 × 3. (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Fundamental Steps in Image Processing

Colour Image Processing: Colours is also as the basis for extracting features of interest in an image



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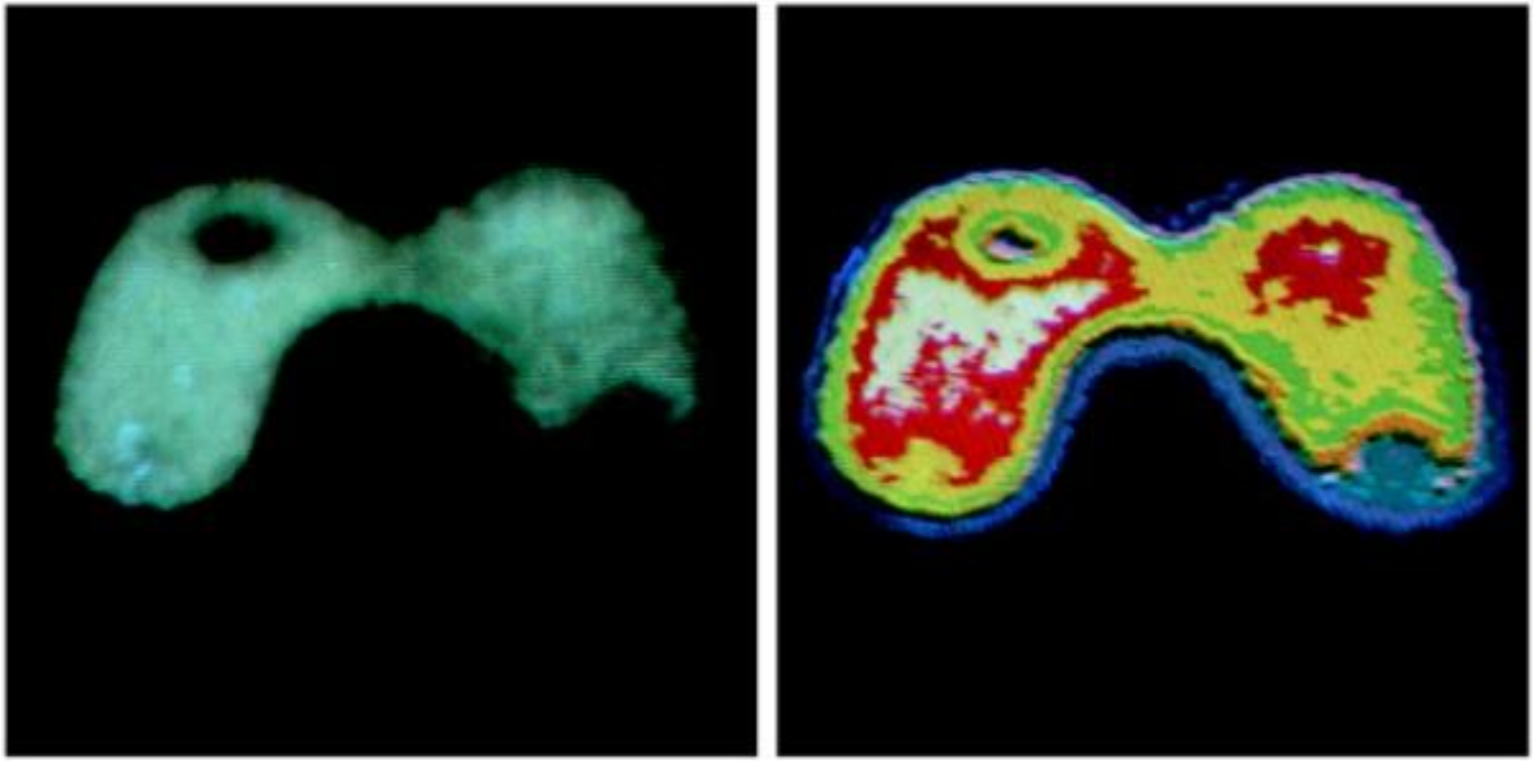
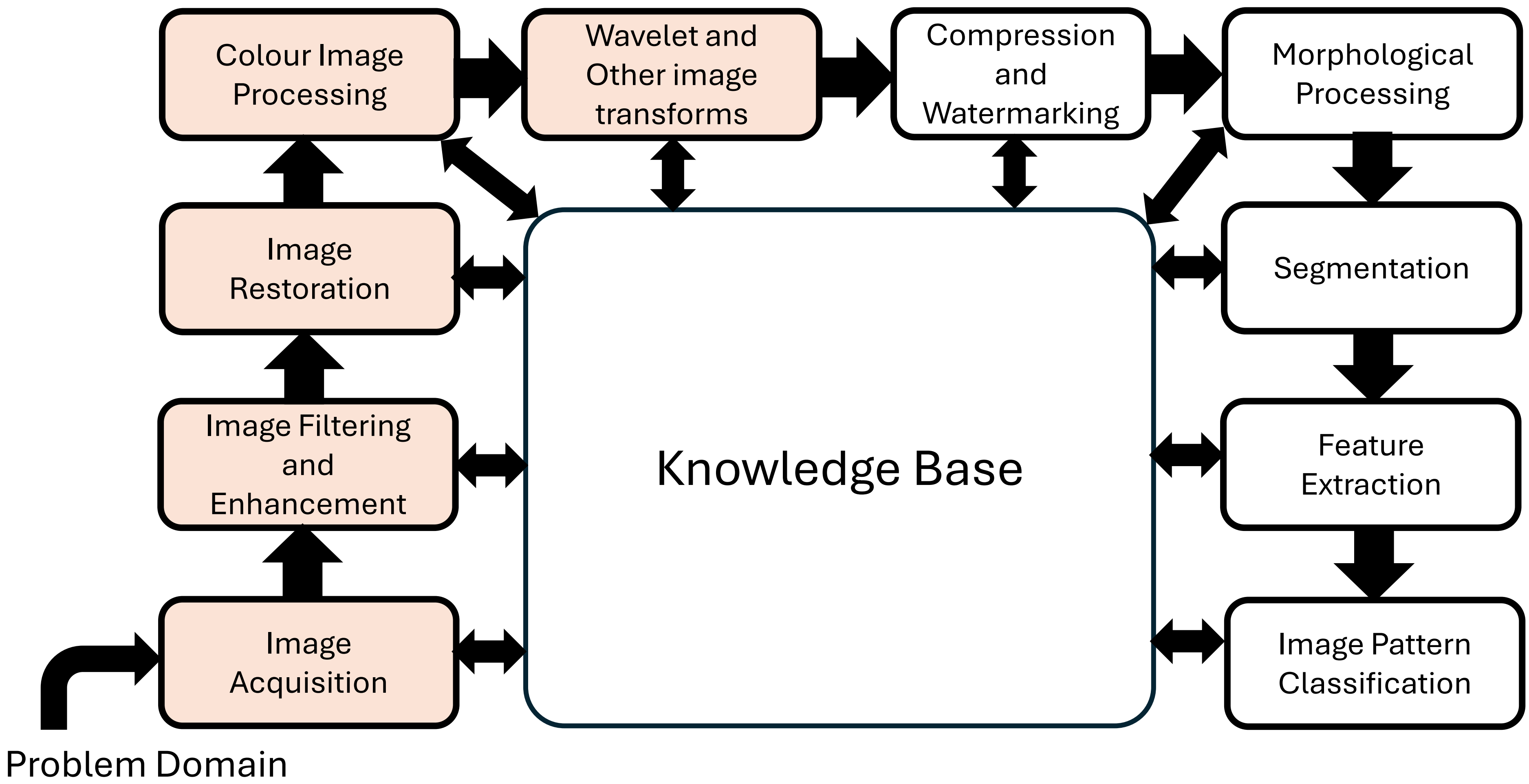


a b

FIGURE 6.20 (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

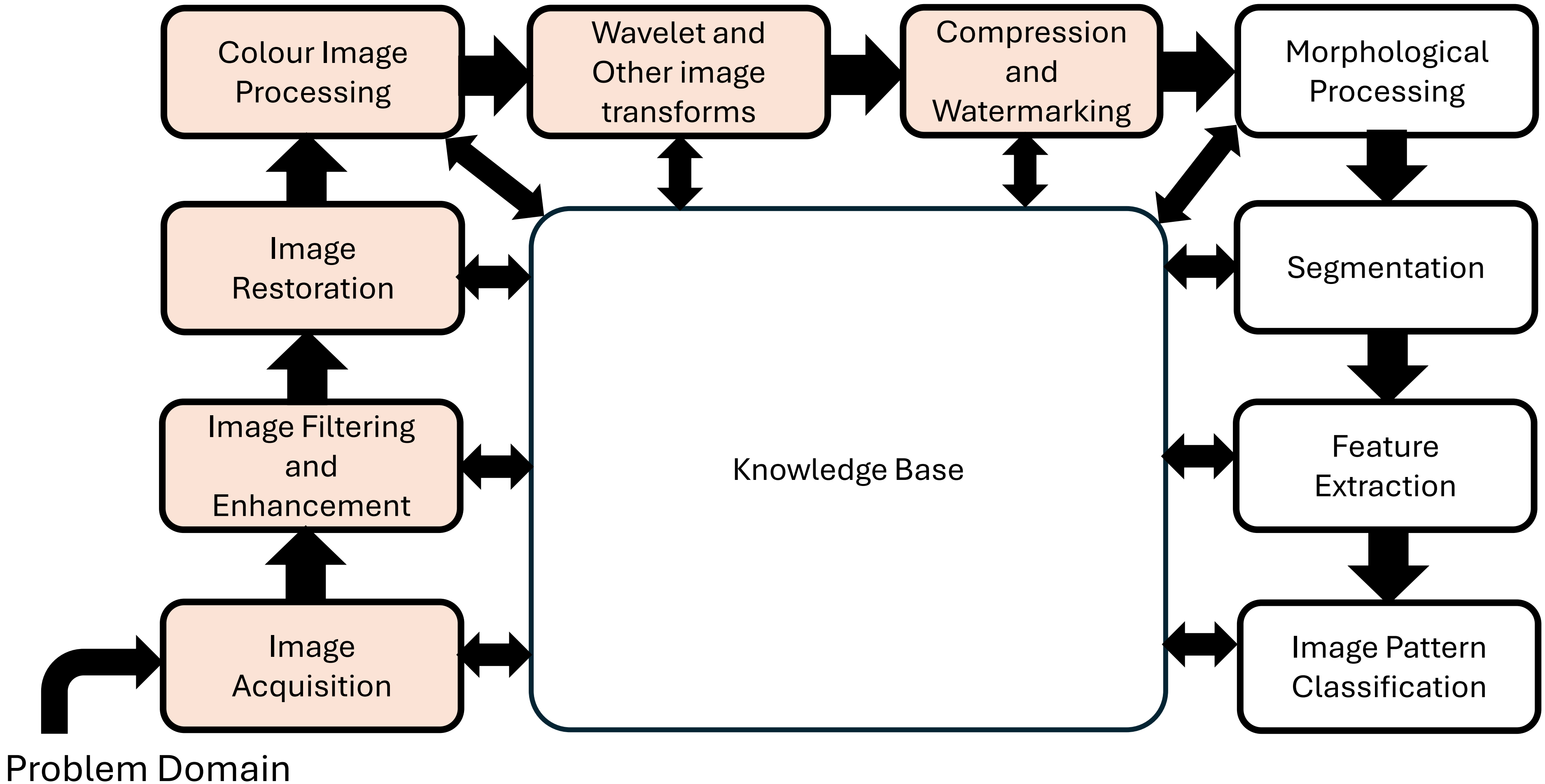
Fundamental Steps in Image Processing

Wavelets: Wavelets are the foundation for representing images in various degrees of resolution.



Fundamental Steps in Image Processing

Compression: It deals with technique to reduce the storage required to save an image or the bandwidth required to transmit it.



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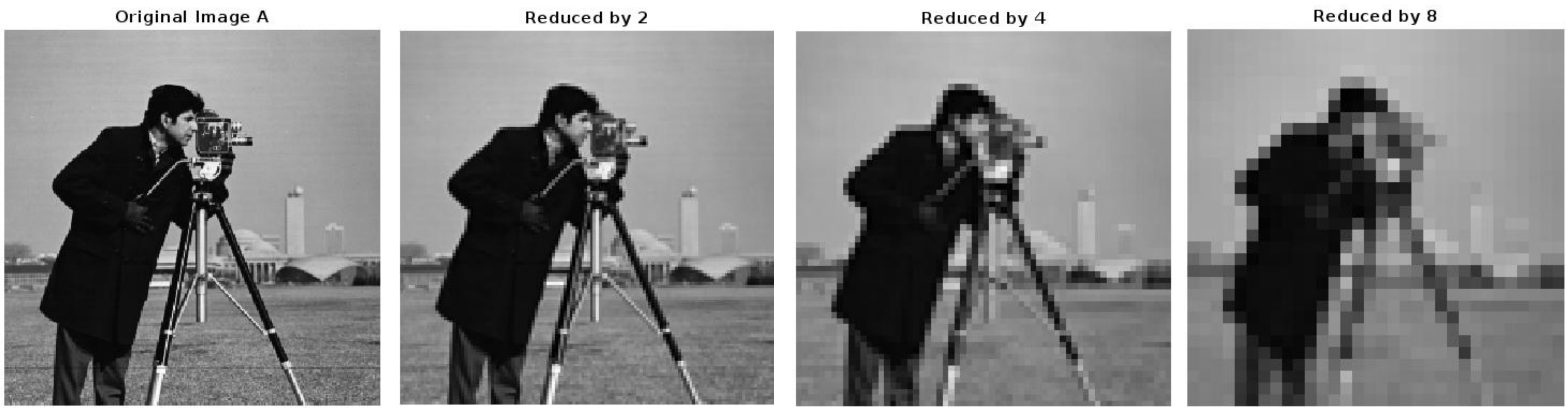
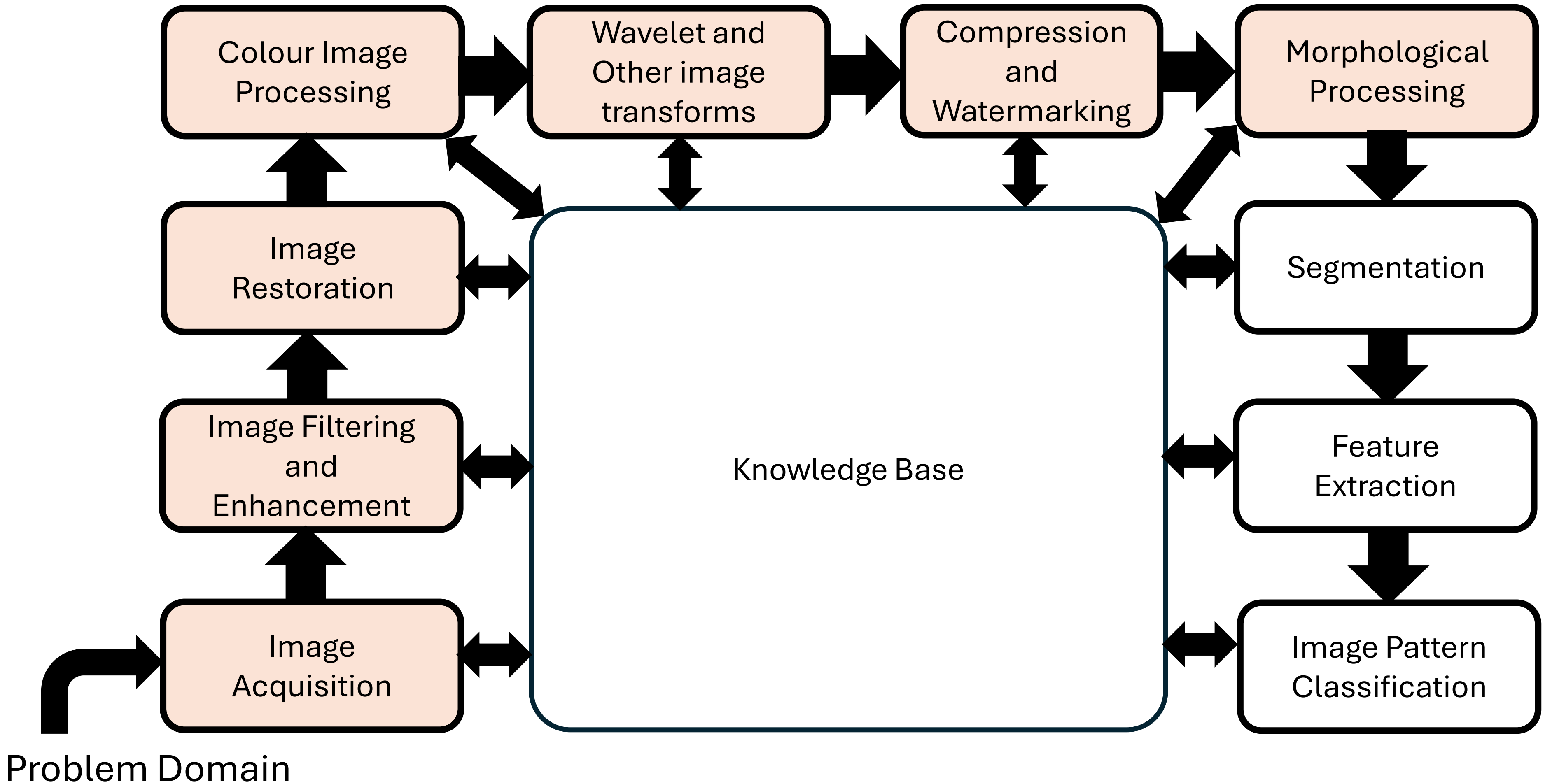


Fundamental Steps in Image Processing

Morphological Processing: It deals with tools for extracting image components that are useful in representation and description of shape.



Fundamental Steps in Image Processing

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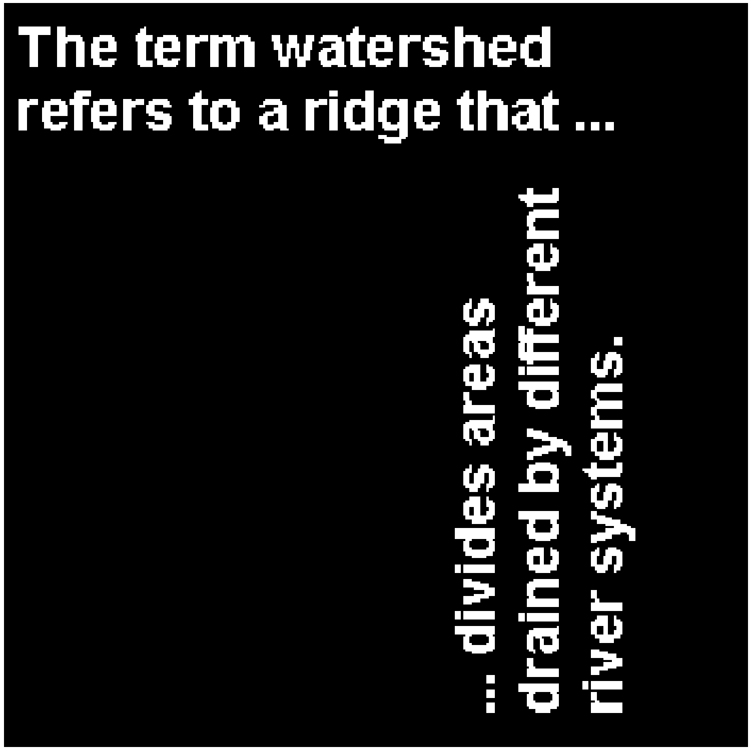
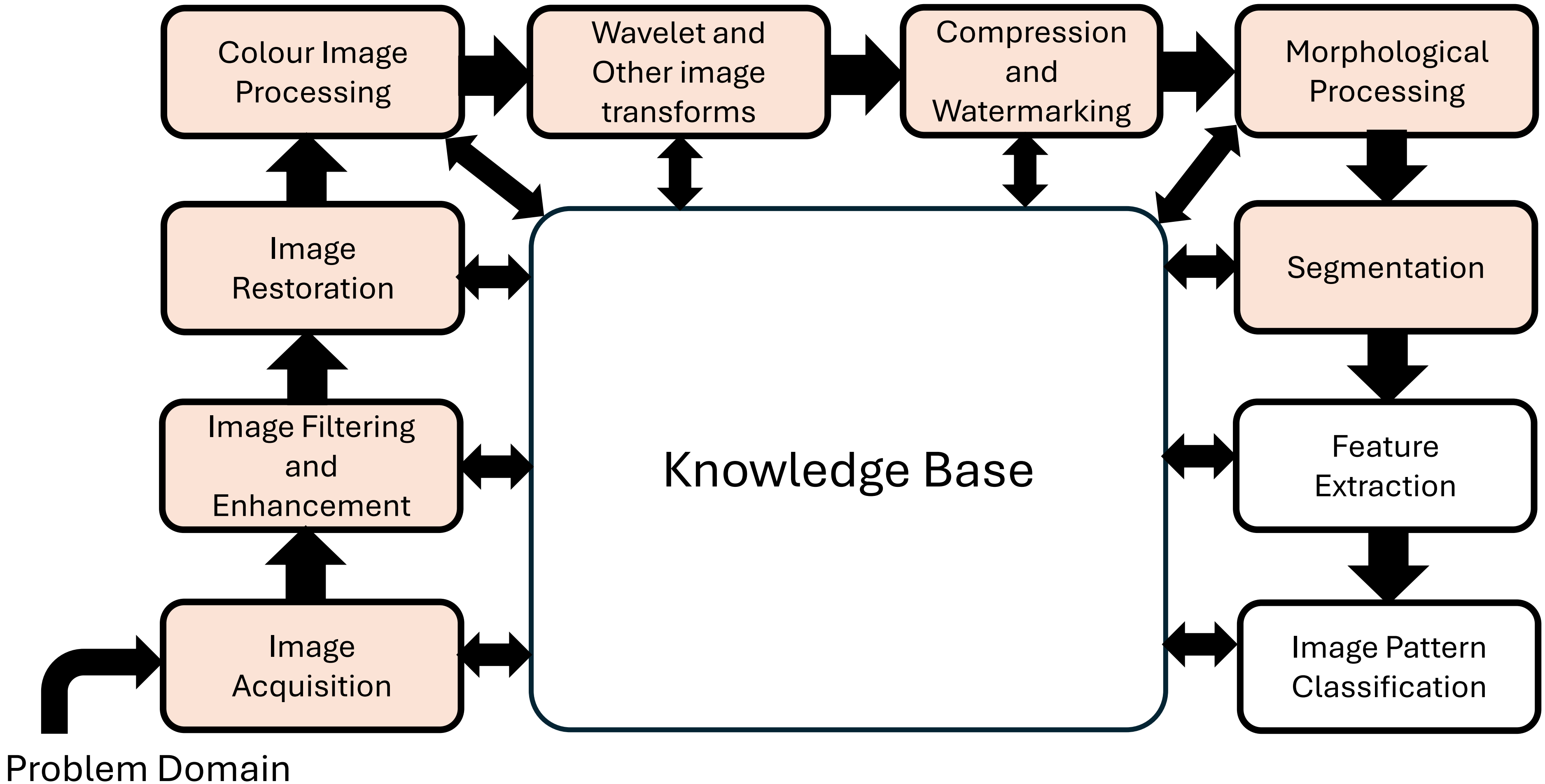
Original text A



External Boundary

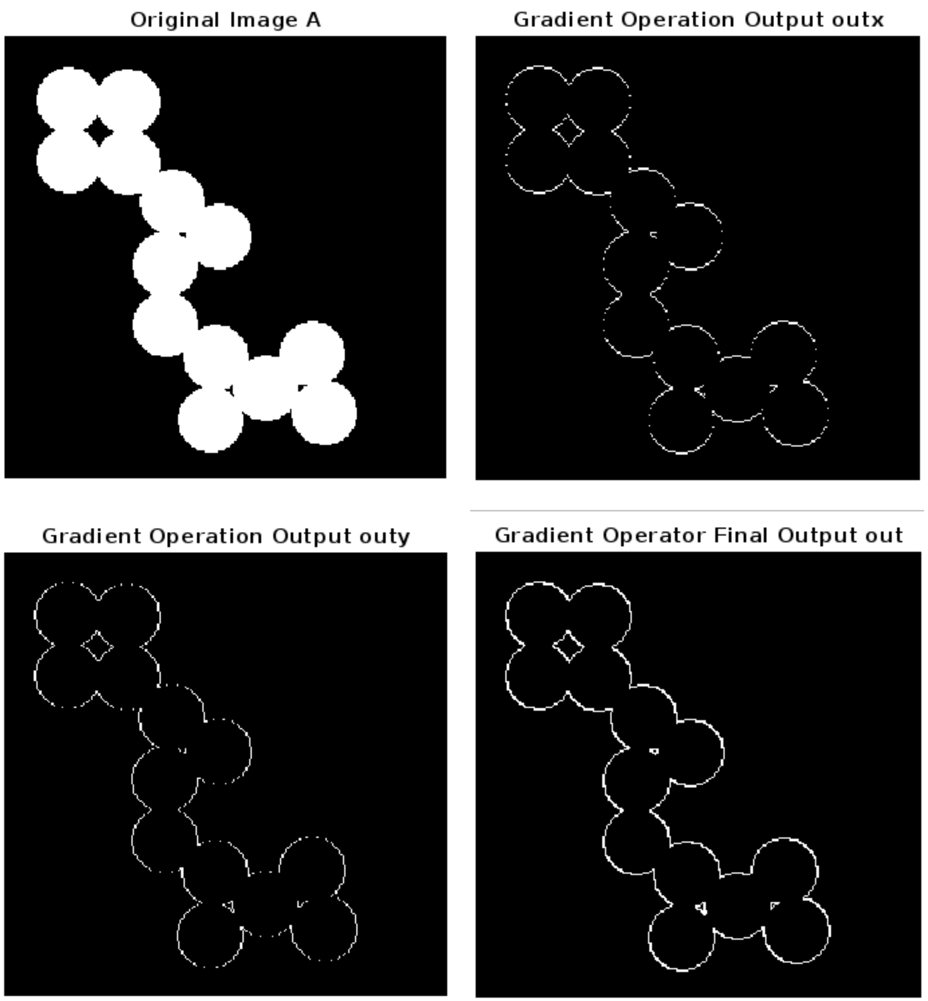
Fundamental Steps in Image Processing

Segmentation: Segmentation partitions an image into its constituent parts of or objects.



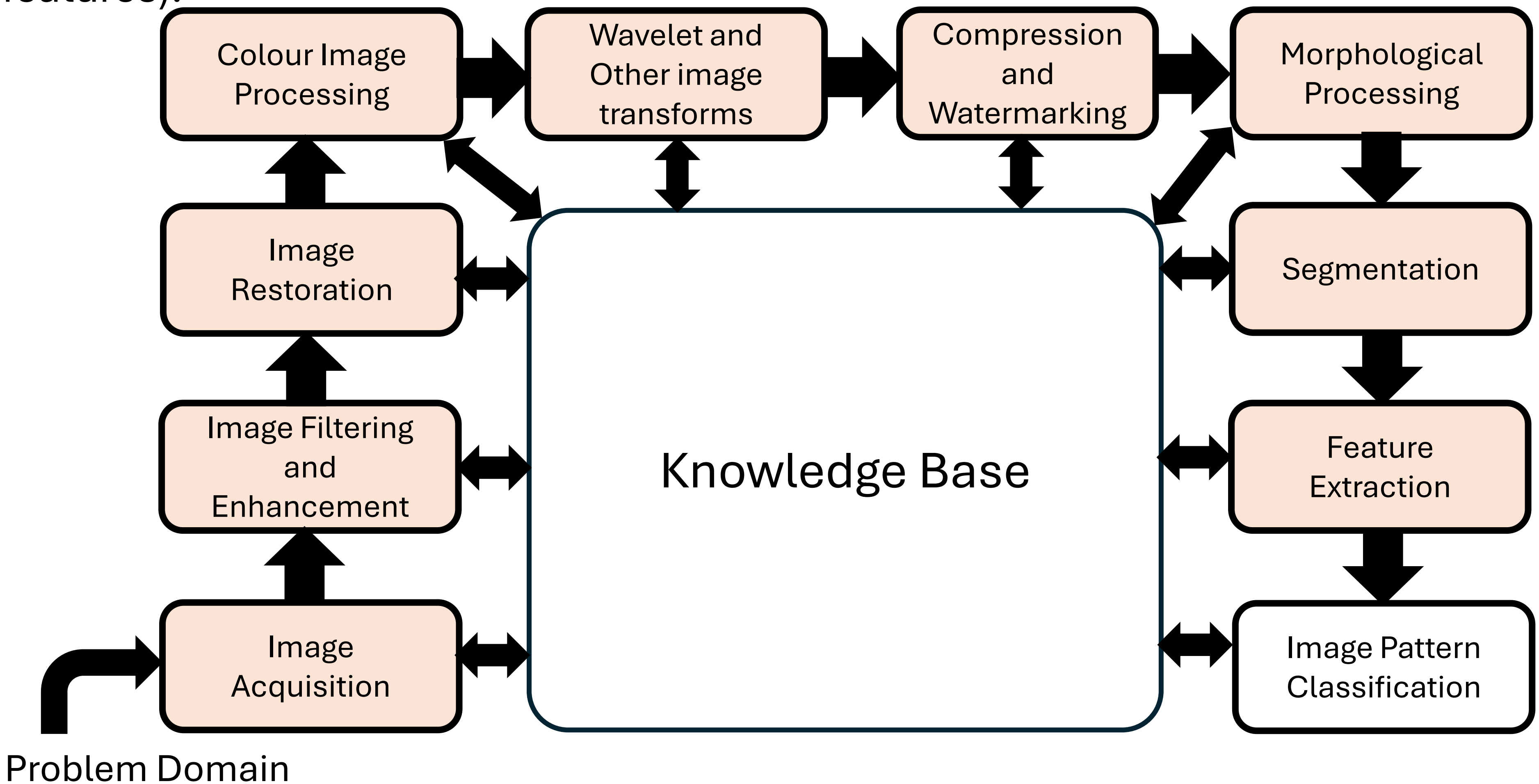
Fundamental Steps in Image Processing

Image Segmentation: Segmentation partitions an image into its constituent parts of or objects.



Fundamental Steps in Image Processing

Feature Extraction: consists of feature detection (finding the region or boundary) and feature description (assigns quantitative attributes to the detected features).



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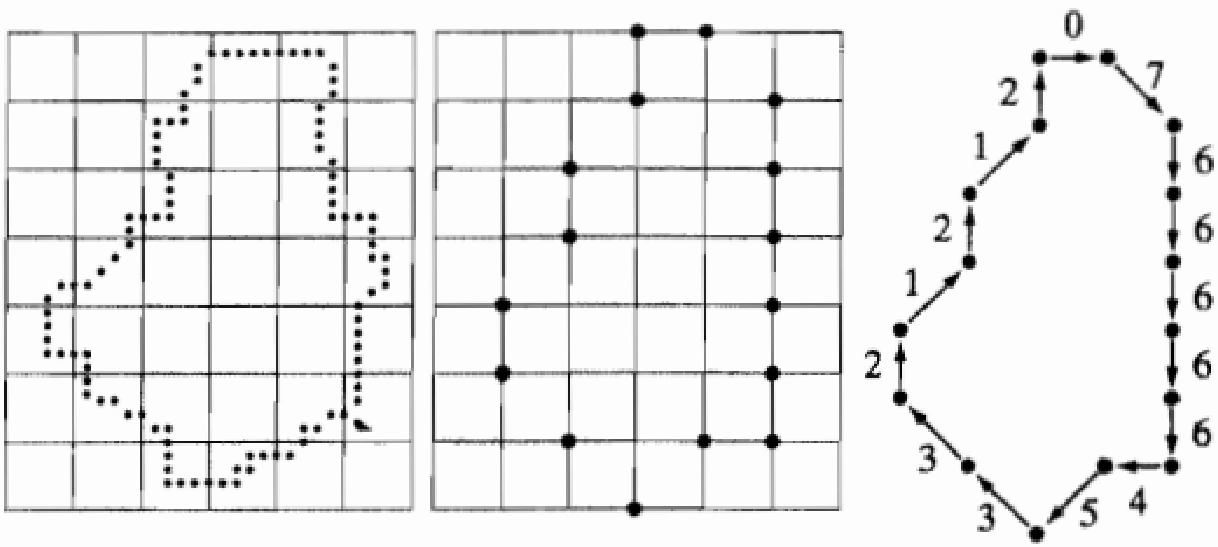
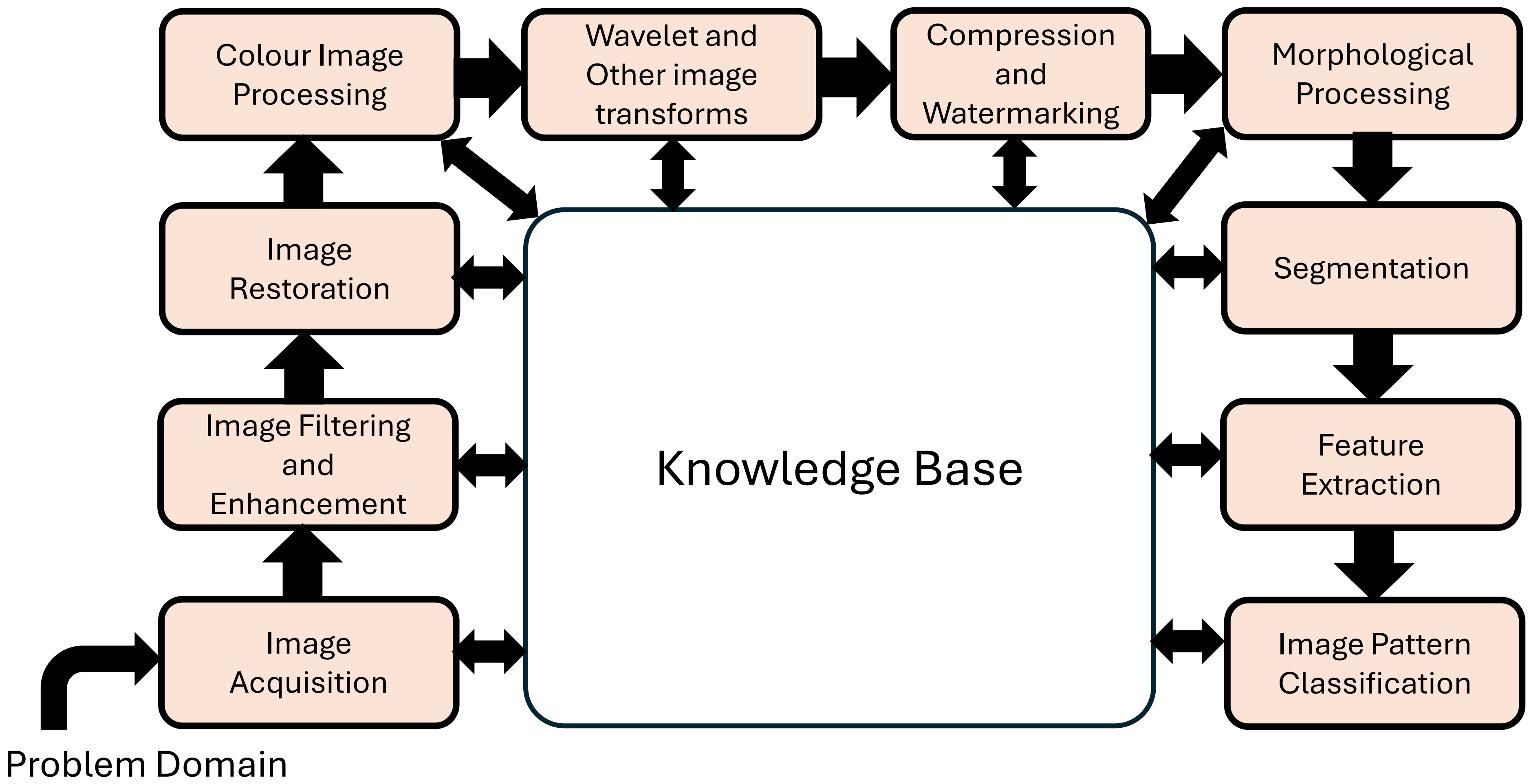


a b c

FIGURE 11.4
(a) Digital boundary with resampling grid superimposed.
(b) Result of resampling.
(c) 8-directional chain-coded boundary.

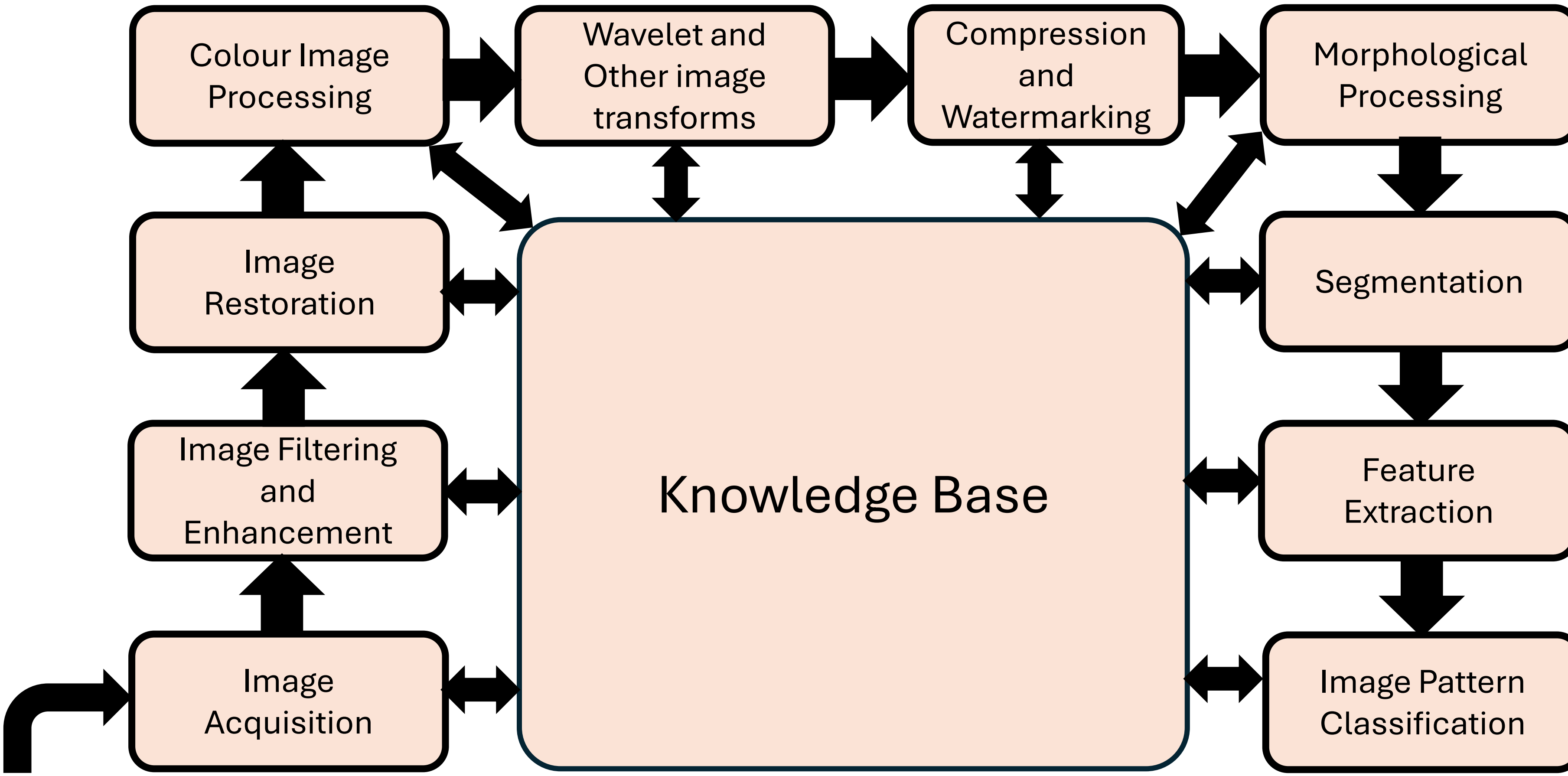
Fundamental Steps in Image Processing

Image Pattern Classification: process that assigns label to an object based on its feature description.



Fundamental Steps in Image Processing

Knowledge Base: Knowledge about a problem domain is coded into an image processing system in the form of knowledge database.



Problem Domain

Thank You